

Kenneth Udovic

Platform / Infrastructure Engineer

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Engineer with 8+ years shipping and operating production systems, focused on the infrastructure and platform layer. Author of **talos-pilot**, a Rust diagnostics TUI for bare-metal Kubernetes on Talos Linux (150+ GitHub stars, shared by Sidero Labs — the makers of Talos), and **AlmanacRAG**, a deployed, fact-grounded retrieval system with Terraform-defined infrastructure and a full evaluation harness. Depth spans cluster internals (etcd quorum, VIP failover, gRPC control-plane APIs), containerization and CI/CD, identity (Keycloak/OIDC), and the full application stack.

OPEN SOURCE & INDEPENDENT PROJECTS

talos-pilot — Author & Maintainer Rust · Talos · Kubernetes

github.com/Handfish/talos-pilot — 150+ stars · shared by Sidero Labs (makers of Talos Linux)

- Built a terminal UI for operating bare-metal Kubernetes on Talos Linux: node monitoring, log streaming, etcd quorum health with member and leader tracking, KubeSpan peers, and diagnostics.
- Wrote a Talos gRPC machine-API client from scratch in Rust (*talos-rs*), surfacing workload health across deployments, statefulsets, and pods, plus a wizard for bootstrapping new clusters on bare metal.
- Fixed multi-node control-plane targeting (VIP and node routing) and merged an outside contributor's fix for VIP request-forwarding failures — real bug reports from operators running the tool on their own clusters.
- Shipped via Homebrew tap, shell and PowerShell installers, and a Nix flake; all cluster operations written to an audit log.

AlmanacRAG — Author TypeScript · PostgreSQL/pgvector · Terraform

github.com/Handfish/AlmanacRAG — live: almanac-rag.pages.dev

- Designed and deployed a fact-grounded RAG over a live 995-page course catalog: the model picks which rows to show and Postgres supplies every price, date, and status, so hallucinated facts are structurally impossible rather than tuned away.
- Built hybrid retrieval — vector kNN and BM25 fused by Reciprocal Rank Fusion in one SQL round trip — behind a typed natural-language-to-filter router, so structured predicates resolve deterministically instead of relying on the model's attention.
- Provisioned all infrastructure as code in Terraform (Neon, Google Cloud Run, Cloudflare Pages) with CI-driven deploys from GitHub Actions; 160+ tests, including Testcontainers with a real Postgres per suite, gated in CI.
- Documented the decisions behind what was deliberately *not* built — no vector index, no graph database, no bitemporal modeling — each grounded in measurements against the live corpus.

EXPERIENCE

Senior Applications Developer — Rutgers University, Division of Continuing Studies Apr 2018 – Present

Lead full-stack developer · sole technical contact for non-technical leadership · runs code review and mentors junior and student developers.

- Deployed and manage a Keycloak instance federated with university identity providers, giving developers OIDC single sign-on against central authentication.
- Introduced containerized deployments and CI/CD using Podman, systemd, GitHub Actions, and Rundeck.
- Stood up multi-server AWS (EC2) hosting on-demand WebRTC classrooms for 400+ attendees on short notice.
- Ran hands-on evaluations of orchestration and infrastructure-as-code tooling — Kubernetes (K3s, Rancher, ArgoCD), Pulumi, SST — to inform deployment strategy.
- Replaced paid vendors with customized open-source deployments, saving the department \$70,000+ per year.
- Architected a Dockerized ETL pipeline that compresses an Oracle database into a binary datafile, enabling performant offline use of a PWA front end.
- Built a Turborepo monorepo (Next.js, Postgres, Keycloak/NextAuth) for an HR onboarding platform — dynamic shared forms, contract generation, DocuSign, CLI tools, cron, and queue systems.
- Delivered high-scale front ends: Elasticsearch-backed paginated search, multimillion-row dataset views, and a D3/Svelte chart-building system; published a React Native app to the iOS and Android stores.

Software Engineer — Fusion Health Aug 2016 – May 2017

- Built full-stack single-page apps integrating GE Healthcare's Centricity EMR for correctional-facility providers; created a templating, publishing, and encryption system for legal EMR documents with digital and fingerprint signatures; scripted pharmaceutical data-migration software for accurate prescription-record transfer.

Technical Intern — Processor Innovations Corporation Jan 2015 – Aug 2016

- Built a mapping application for calculating and graphing antenna signal strength; performed business-intelligence data analysis; built, repaired, and configured Linux servers.

SKILLS

Infrastructure & DevOps Kubernetes (Talos, K3s, Rancher, ArgoCD), Docker / Podman, systemd, GitHub Actions, Rundeck, Terraform, Pulumi, SST, AWS (EC2/ECS/EKS/Lambda), Google Cloud Run, Neon, Cloudflare Pages, Nginx / Caddy / Apache, Keycloak / OIDC

Languages TypeScript, Rust, PHP, C#, Bash, SQL

Backend & Data Node.js (Effect, Hono, Express), Laravel, PostgreSQL / pgvector, MySQL, SQLite, MS SQL Server, Elasticsearch, Redis, REST, WebSockets

Frontend React / React Native, Next.js, Astro, Svelte, Vue, Tailwind CSS, Radix UI, Storybook, Cypress

Practices Monorepos (Turborepo), microservices, CI/CD, infrastructure as code, evaluation harnesses, observability

EDUCATION

B.S. Computer Science, Minor in Entrepreneurship May 2016

Rutgers University – New Brunswick · GPA 3.4